

M18 - Project Quick Reference

USE DURING BUILD / REVISION

1. Grain

Write: one row represents _____. Define business key before joins/aggregations.

2. Reconciliation

source = trusted + quarantined + deliberately deduplicated/filtered outcomes under stated rules.

3. Raw

Preserve received source evidence + batch metadata. Do not rewrite raw to make it "clean."

4. Quality gate

Critical bad data blocks trusted publish or is quarantined according to explicit rule.

5. Idempotency

Repeat same logical input/work; trusted logical counts/totals must remain correct.

6. Incremental

State unique key + new/changed-row filter/watermark + late-data behavior.

7. Backfill

Bound a historical range, reuse safe/idempotent logic, then revalidate downstream state.

8. Failure

Save first error -> root cause -> smallest repair -> rerun failed path -> full controls.

9. Orchestration

Dependencies/retries must not turn partial failure into false fresh publish.

10. Security

No real secret in Git/logs; identify PII; least privilege; synthetic public data.

11. Performance

Measure/inspect before optimizing. Tiny local data does not prove production scale.

12. README

Problem -> architecture -> run -> grain/contracts -> quality -> rerun/backfill -> failure -> limitations.

13. Defense

Requirement -> decision -> trade-off -> evidence -> limitation.

14. AI

Use it to explain/review/challenge; do not submit logic you cannot reproduce/defend.